Using Tree-Based Reinforcement Learning Methods to Support Personalized Decision-Making in Hand Treatment.

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CLINICAL & AGENTIC AI TAKEAWAY

Tree-based reinforcement learning methods can optimize personalized hand treatment decisions by balancing competing clinical priorities and enhancing patient-centered care.

ABSTRACT

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: Department of Biostatistics, School of Public Health, University of Michigan, Ann Arbor, 1415 Washington Heights, Ann Arbor, MI 48109, USA. Keywords: Reinforcement Learning, Personalized Healthcare, Adaptive Dynamic Treatment, Hand Surgery Decision-making, Tree-based Decision Optimization Extended Abstract from Repository: Personalized treatment enhances healthcare by tailoring optimal decisions to each patient based on their specific characteristics and treatment history. Reinforcement learning (RL) methods are powerful tools for estimating optimal, data-driven, dynamic treatment decision rules. This article presents a tutorial on Tree-based RL and Multi-Objective Tree-based RL for advancing the estimation of optimal dynamic treatment regimes. Data from the Silicone Arthroplasty in Rheumatoid Arthritis study demonstrate their application in optimizing joint arthroplasty decisions. These methods support personalized, data-driven strategies while balancing competing clinical priorities, aiding clinicians in making informed, patient-centered decisions within ethical and practical constraints.

Keywords: Reinforcement Learning, Personalized Healthcare, Adaptive Dynamic Treatment, Hand Surgery Decision-making, Tree-based Decision Optimization

Study Overview & Background

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joint arthroplasty decisions. These methods support personalized, data-driven strategies while balancing competing clinical priorities, aiding clinicians in making informed, patient-centered decisions within ethical and practical constraints.

Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Tree-based reinforcement learning methods can optimize personalized hand treatment decisions by balancing competing clinical priorities and enhancing patient-centered care.

Extended Repository Abstract

Personalized treatment enhances healthcare by tailoring optimal decisions to each patient based on their specific characteristics and treatment history. Reinforcement learning (RL) methods are powerful tools for estimating optimal, data-driven, dynamic treatment decision rules. This article presents a tutorial on Tree-based RL and Multi-Objective Tree-based RL for advancing the estimation of optimal dynamic treatment regimes. Data from the Silicone Arthroplasty in Rheumatoid Arthritis study demonstrate their application in optimizing joint arthroplasty decisions. These methods support personalized, data-driven strategies while balancing competing clinical priorities, aiding clinicians in making informed, patient-centered decisions within ethical and practical constraints.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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